Sydney, NSW
dhruvlawaniya2510@gmail.com

Dhruv Lawaniya

ML Engineer / Data Scientist

github.com/DhruvLawaniya
linkedin.com/in/dhruvlawaniya

Summary

Master of Computer Science candidate at the University of Sydney , specializing in Machine Learning and Data Science. Experienced in designing scalable data pipelines, building production-ready ML models, and delivering enterprise analytics solutions using Python, Spark, and SQL.

Skills

Programming: Python, Scala, SQL, Java, C/C++
Machine Learning: Scikit-learn, Model Evaluation, Feature Engineering, NLP, Computer Vision
Data Engineering: Apache Spark, Spark-SQL, Data Pipelines
Data Visualization: Tableau, Power BI, Matplotlib, Seaborn
Tools: Git, Jupyter Notebook, VS Code
Databases: MySQL

Work Experience

Data Scientist **05/2023 – 05/2024**
Deloitte India

- Designed and deployed scalable Scala-Spark data pipelines to process large-scale audit datasets across 100+ enterprise clients, improving processing efficiency and ensuring 100% compliance accuracy.
- Built Python-SQL analytics frameworks to automate anomaly detection and large dataset validation, reducing manual audit effort and improving turnaround time.
- Developed executive-level dashboards using Tableau and Power BI to translate complex analytical outputs into actionable business decisions.
- Collaborated with cross-functional stakeholders to convert regulatory and business requirements into production-ready analytics systems under strict deadlines.

Education

Master of Computer Science (Advanced Entry) **02/2025 – Present**
University of Sydney, Sydney, Australia

- Specialisations: Data Science and AI, Cybersecurity
- WAM: Distinction(81)

Bachelor of Technology – Computer Science Engineering, Bennett University **2020 – 2024**
Greater Noida, India *CGPA: 9.16*

Projects

ATIS Natural Language to SQL System

- Built an end-to-end Natural Language to SQL system using the ATIS (Air Travel Information System) dataset to translate user questions into executable SQL queries.
- Developed and evaluated multiple machine learning approaches including feedforward neural networks, sequence tagging, and large language model prompting (zero-shot, few-shot, and many-shot).
- Implemented data preprocessing, feature engineering, intent classification, and variable extraction pipelines using Python, PyTorch, and Scikit-learn.
- Achieved reliable SQL query generation through model evaluation, hyperparameter tuning, and comparison of neural network and LLM-based approaches.

Jade – ML-Based Home Gardening Assistance Platform

- Trained and evaluated computer vision models for plant disease detection, optimizing performance via hyperparameter tuning.
- Built backend services using Node.js to integrate ML inference pipelines into a production-ready web application.

Extra Curricular

PG Connect Advisor
University of Sydney

- Assisted postgraduate students with academic transition, networking, and campus integration.

USU Volunteer
University of Sydney

- Supported university-wide events and student engagement initiatives.